frontiers | Frontiers in Education

TYPE Systematic Review
PUBLISHED 29 January 2026
DOI 10.3389/feduc.2025.1723188

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OPEN ACCESS

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RECEIVED 11 October 2025
REVISED 08 December 2025
ACCEPTED 12 December 2025
PUBLISHED 29 January 2026

CITATION
Takeuti AMN (2026) Learning outcomes improvement in primary and lower secondary education: a systematic review of education reform policies on a state-level case in Brazil.
Front. Educ. 10:1723188.
doi: 10.3389/feduc.2025.1723188

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Learning outcomes improvement in primary and lower secondary education: a systematic review of education reform policies on a state-level case in Brazil

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This study examines the education reform implemented in the State of Ceara, Brazil, which has achieved notable improvements in learning outcomes, mainly in primary and lower secondary education, despite persistent socioeconomic constraints. The research aims to identify the policies that explain the continuity and effectiveness of the Ceara reform over time. A qualitative approach was employed, following the PRISMA 2020 reporting guidelines. The review was guided by two research questions: (1) How can academic studies on learning improvements in Ceara be characterized; and (2) Which components of Ceara's education policies are associated with reported improvements in learning outcomes. Eighty studies published between 2005 and 2024 were selected from academic databases and coded to identify policy instruments, governance arrangements, and implementation dynamics associated with improved learning outcomes. The findings reveal that Ceara's education model emerged from the convergence of coherent governance, performance-based financing, and sustained teacher training that reinforced one another over time. The reform was marked by institutional continuity, strong collaboration between state and municipal governments, and the use of assessments, as well as incentives to guide decision-making. However, the review also highlights risks associated with competitive funding and test-driven accountability, including exclusionary practices and a limited focus on measurable learning dimensions. Overall, the analysis demonstrates that Ceara's education reform exemplifies how multi-level policy coherence and long-term institutional commitment can yield measurable learning outcomes, while also highlighting the importance of striking a balance between efficiency, inclusion, and educational equity.

KEYWORDS

education reform, education policies, learning outcomes, primary education, lower secondary education, Ceara state

1 Introduction

Education is increasingly recognized as a central dimension in addressing global challenges and promoting more democratic, responsible, meaningful, and equitable livelihoods (Wals, 2015). Within the Sustainable Development Goals, SDG 4 emphasizes the need to ensure inclusive and equitable quality education and promote lifelong learning

opportunities for all (United Nations, 2015; UNESCO, 2015a,b; UNESCO, 2017). Despite this recognition, many developing countries continue to struggle with converting expanded enrollment and increased spending into measurable improvements in student learning outcomes (World Bank, 2018). Learning outcomes refer to the results that students achieve following a learning process. Learning outcomes are commonly expressed through quantifiable measures such as grades or standardized test scores, which serve as indicators of instructional effectiveness and the extent to which educational objectives have been achieved (Kamalia et al., 2022).

The persistent learning crisis is evident in low- and middle-income countries, where students, particularly those from lower socioeconomic backgrounds, tend to perform worse compared to their peers from wealthier areas (Antoninis et al., 2016). Ensuring inclusivity and equality in education, leaving no one behind, remains an urgent issue for the global education agenda (Boeren, 2019). Brazil offers a pertinent context for examining these challenges. The country operates under a decentralized educational system in which the federal government, states, the Federal District, and municipalities share responsibility for ensuring universal access, quality, and equity in compulsory education (Brasil, 1988). Within this framework, the federal government is responsible for establishing national education guidelines, financing redistribution policies, and providing support to states and municipalities. States are tasked with coordinating their respective education systems, particularly at the secondary level, and providing technical and financial assistance to municipalities. Municipalities, in turn, hold the main responsibility for managing early childhood and primary education. Although this collaborative structure has facilitated the expansion of access to basic education over the last almost forty years, student performance continues to lag, and significant inequalities persist (de Oliveira and de Araujo, 2005; Marchelli, 2010; Bruns et al., 2012; Guimaraes et al., 2024). Socioeconomic, racial, and disability factors contribute to low achievement, and regional disparities in learning outcomes mirror differences in socioeconomic conditions. While southern and southeastern states generally meet or surpass national averages on standardized assessments, northern and northeastern states, as well as black children and students from lower-income families, tend to score below the national average (Instituto Nacional de Estudos e Pesquisas Educacionais Anisio Teixeira (INEP), 2023).

Ceara has drawn national and international attention for its remarkable educational progress in primary and secondary education despite adverse socioeconomic conditions. Located in Brazil's northeast region and home to approximately nine million inhabitants, the state ranks among the 27 with the five lowest gross domestic product per capita, at around USD 4,300 in 2023 (Instituto Nacional de Estudos e Pesquisas Educacionais Anisio Teixeira (INEP), 2023; IBGE, 2023). Nevertheless, Ceara has achieved learning outcomes in large-scale assessments that surpass those of wealthier states (Loureiro and Cruz, 2020; Loureiro et al., 2020). Since 2007, its municipalities have shown the greatest improvements in national standardized assessments such as Prova Brasil, consistently outperforming richer regions in the Basic Education Development Index (IDEB) (Loureiro et al., 2020). Between 2007 and 2019, according to the National Literacy Assessment (ANA), literacy rates among second-grade students rose steadily, narrowing historical achievement gaps (Instituto Nacional de Estudos e Pesquisas Educacionais Anisio Teixeira (INEP), 2023). By 2023, several municipalities in Ceara ranked in leading positions in the national IDEB rankings for primary education. In Ceara, the municipal education system holds a substantial share of primary enrollments. In 2007, municipal schools accounted for approximately 98% of public primary enrollments in the state, compared with about 76% nationally (Loureiro et al., 2020; World Bank, 2020). This concentration of municipal responsibility is not equally present in all Brazilian states, reflecting the institutional complexity and coordination challenges involved in implementing state-level support mechanisms.

Recognized as a "model of educational success" (Abrucio et al., 2020; Loureiro and Cruz, 2020; Loureiro et al., 2020), Ceara is often cited as a positive deviant in the Brazilian educational landscape (Instituto Nacional de Estudos e Pesquisas Educacionais Anisio Teixeira (INEP), 2023). Its education policies have inspired federal and state-level programs, with the Brazilian Federal Government and other states currently borrowing components of Ceara's reform to their own education systems (Brazil, 2013, 2023; Todos pela Educação, 2021). Its experience provides a valuable case for examining coordinated intergovernmental collaboration in countries with sub-national education systems, which justifies conducting a systematic literature review to consolidate and critically assess existing evidence on the mechanisms and outcomes of Ceara's educational reform.

1.1 Education reform, policy implementation, and learning outcomes improvements in developing contexts

Education reform has emerged as a central strategy to address persistent challenges in improving learning outcomes across developing countries. The global education agenda has increasingly shifted from emphasizing access to schooling toward prioritizing educational quality, equity, and measurable learning outcomes (UNESCO, 2022; World Bank, 2018). In many low- and middle-income countries, expanding enrollment has not necessarily translated into improvements in student learning, a condition often described as a "learning crisis" (Pritchett, 2013). Learning poverty, the inability of a child to read and understand a simple text by the age of ten, illustrates the urgent need for systemic transformations that strengthen both instructional quality and governance structures. Within this context, education reforms represent deliberate and fundamental changes in the structures, systems, or practices of educational institutions designed to enhance learning effectiveness (Kim, 2019; Roofe, 2014). Such reforms often aim to improve curriculum design, school governance, teaching practices, assessment methods, or resource allocation to meet evolving societal and economic demands. Empirical evidence demonstrates that reforms directly linked to classroom practice yield the most significant and consistent gains in student achievement in developing countries (Kremer et al., 2013; McEwan, 2015; Glewwe and Muralidharan, 2015; Angrist et al., 2023).

Education reforms are implemented through education policies, which serve as the mechanisms for change. These policies provide the structure through which reforms are designed,

executed, and evaluated. Interventions solely focused on resource provision, such as textbook distribution, class size reduction, or unconditional teacher salary increases, have shown limited or no effects on learning outcomes unless integrated into broader systemic reforms that address teacher quality, accountability, and pedagogical support (Glewwe and Muralidharan, 2015). As noted by Hanushek (2011), Bruns and Luque (2015), and Schleicher (2018), the effectiveness of education reform depends on the strategic alignment between policy decisions and evidence-based implementation, particularly in domains related to teacher performance, leadership, and monitoring of results.

Education policies can be defined as "actions taken by governments in relation to educational practices and how governments address the production and delivery of education in a given system" (Viennet and Pont, 2017, p. 19). They establish the institutional framework through which reforms are operationalized, guiding processes of decision-making, coordination, and accountability. Therefore, education reforms and policies are intrinsically connected. While reforms represent the strategic transformations intended to improve education systems, policies provide the legal, administrative, and operational instruments necessary to enable and sustain such transformations.

In the field of public policy, analyzing priority setting within education policy is essential. Policy factors refer to the conditions, elements, or decisions that influence the formulation, implementation, and impact of public policies. Within this scope, policy priorities, defined as the most important ideas or plans guiding decision-making and resource allocation (Collins, 2023), play a determining role. Since time, funding, and institutional capacity are limited, prioritizing actions and decisions becomes critical for achieving reform goals. Effective prioritization ensures that limited resources are strategically allocated to initiatives with the highest potential to improve learning outcomes. This rationale aligns with Hanushek's (2011) argument that efficient resource management and teacher performance are fundamental determinants of educational quality.

In developing countries such as Brazil, analyzing how policies are prioritized gains particular relevance due to persistent regional inequalities in educational performance. This article examines the specific education policies and related factors implemented in Ceara that have been associated with improvements in primary and lower secondary education learning outcomes. It includes the characteristics of the academic studies reviewed and presents the central education policies that composed Ceara's education reform.

1.2 Research gaps

Despite the international relevance of systemic reforms for improving learning outcomes, there remains a need to investigate how specific subnational experiences, such as Ceara's in Brazil, translate theoretical principles into effective policy design, implementation, and sustainability (Carnoy et al., 2017). Although Ceara's case has been widely studied, most research has focused on isolated policy measures, without integrating these findings into a comprehensive account of how multiple policy factors interacted to sustain reform over time (Crouch, 2020). Existing studies offer valuable insights into specific components of Ceara's education reform. However, there is a lack of synthesis that can explain how these diverse elements work together to produce consistent and long-term progress in student achievement.

Furthermore, the diffusion of Ceara's model to other Brazilian states has not consistently yielded comparable results, underscoring the importance of analyzing the contextual and institutional conditions that enabled its success (Todos pela Educação, 2021; Venâncio, 2020). While Ceara's policies have often been portrayed as a replicable model for educational improvement, the variation in outcomes across states suggests that transferability depends on specific local arrangements, leadership capacity, and political commitment.

Debates within Brazil also reflect tensions surrounding Ceara's education reform, with distinct political and institutional actors framing its outcomes through divergent ideological lenses, ranging from neoliberal critiques emphasizing managerial efficiency to narratives highlighting democratic participation and collaborative governance (Spring, 2015; UNESCO, 2021; Venâncio, 2020). This ideological diversity points to the need for research that can navigate these interpretations and assess reform outcomes through empirically grounded and methodologically consistent frameworks.

Additionally, Loureiro et al. (2020) emphasize the need to adopt systematic and comprehensive approaches to document education reforms effectively, as much of the existing literature focuses on outcomes without adequately linking them to the specific strategies and implementation processes employed. This concern is echoed by McNaught (2022) and Ponne (2023), who argue that the scarcity of structured analyses undermines both policy evaluation and the capacity to inform future reform initiatives. Beyond the regional scope, comparative studies in developing countries reinforce similar analytical gaps. The World Bank's (2023) synthesis of education research highlights that many low- and middle-income countries continue to struggle with learning crises exacerbated by COVID-19, inadequate early childhood education, teacher shortages, and the lack of evidence-based policymaking.

This paper addresses these research gaps by consolidating and reorganizing the existing evidence on Ceara's education policies, institutional arrangements, and governance practices. Accordingly, the following research questions guided this review: (RQ1) How can academic studies on learning improvements in Ceara be characterized regarding their temporal trends, methodologies, analytical levels, stakeholder engagement, and institutional and geographical distribution? (RQ2) Which components of Ceara's education policies are associated with reported improvements in learning outcomes? To address these research questions, this study conducted a systematic literature review following PRISMA 2020 standards (Page et al., 2021), enabling the transparent identification, screening, eligibility appraisal, and synthesis of studies examining Ceara's education reform model and its reported influence on primary and lower secondary education outcomes. This methodological approach enables the systematic examination of how Ceara's educational policies and reform strategies have been interpreted and evaluated in the literature.

2 Materials and methods

This study adopted a systematic literature review design to synthesize peer-reviewed research on Ceara's education reform and its reported influence on learning outcomes. The review adhered to the PRISMA 2020 reporting guideline, which establishes standards for transparency, replicability, and methodological rigor across all phases of systematic evidence synthesis. Because the screening procedures were initiated prior to the incorporation of PRISMA into this study, a retrospective protocol was adopted. The study selection process followed the PRISMA 2020 guidelines, as illustrated in Figure 1. All methodological decisions and procedural sequences are therefore detailed below to ensure full traceability and compliance with best practices.

The review was guided by two research questions. The first examined how the scholarly literature characterizes the education reform model implemented in Ceara since 2005. The second investigated which components of Ceara's education policies have been identified as influencing learning outcomes across Brazilian states. To ensure conceptual clarity and methodological coherence, both research questions were operationalized through the PICOS framework. In this review, the population corresponds to primary and lower secondary education in the Brazilian state of Ceara, including its schools, students, teachers, and administrative authorities. The interventions refer to the policies, programs, and governance mechanisms associated with Ceara's education reform that the literature links to learning improvements. The comparators consist of implicit contrasts internal to the state, such as different phases of policy implementation, student cohorts, or municipal arrangements, as reported in the analyzed studies; no external state-level comparators were required at this stage of the review. The outcomes include reported effects on learning indicators, literacy levels, school performance measures, and accountability or governance arrangements in basic education. Finally, the study designs encompass empirical, theoretical, descriptive, and mixed-methods peer-reviewed research published between 2005 and 2024 that examine Ceara's education policies and their relationship with improvements in learning outcomes. Table 1 presents the operationalization of the research questions using the PICOS framework.

The search strategy was conducted in three databases that host peer-reviewed journals and multidisciplinary research outputs relevant to educational studies in Brazil. Google Scholar served as the first search environment due to its comprehensive coverage of global academic literature indexed from universities, publishers, and scientific repositories. Capes Periódicos, maintained by the Coordination for the Improvement of Higher Education Personnel (Capes), provided structured access to Brazilian and international journals in education, public policy, and related fields, ensuring

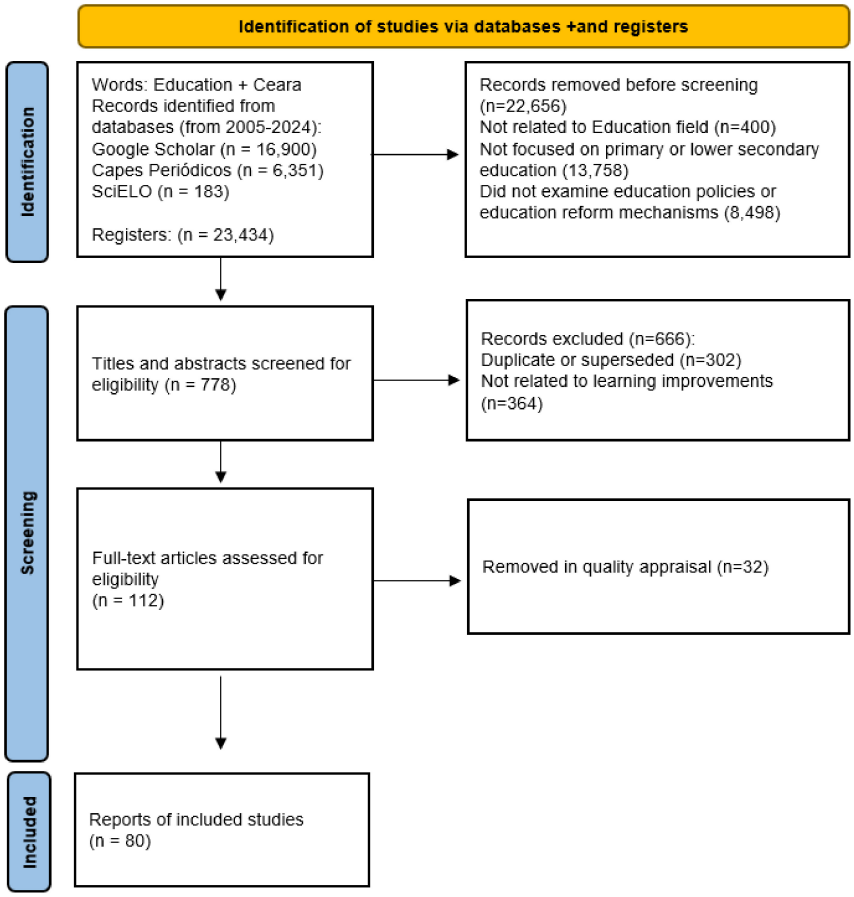


FIGURE 1
PRISMA 2020 flow diagram of the selection process for studies on Ceara education.

TABLE 1 Operationalization of the research questions using the PICOS framework.

PICOS element	Operational definition in this review
Population (P)	Primary and lower secondary education in the Brazilian state of Ceara, including its schools, students, teachers, and administrative authorities.
Interventions (I)	Policies, programs, and governance mechanisms associated with Ceara's education reform that the literature identifies as contributing to learning improvements.
Comparators (C)	Implicit temporal and contextual contrasts within Ceara, such as different phases of the reform, student cohorts, or municipal implementation contexts reported by the studies; no external state-level comparators were required at this stage.
Outcomes (O)	Reported effects on learning indicators, literacy levels, school performance measures, and accountability or governance arrangements associated with basic education in Ceara.
Study designs (S)	Empirical, theoretical, descriptive, and mixed-methods peer-reviewed studies published between 2005 and 2024 that examine Ceara's education policies and their relationship with improvements in learning outcomes.

the inclusion of research not indexed elsewhere. SciELO (Scientific Electronic Library Online), an open-access platform with strong representation of Latin American scholarship, was included due to its mission of enhancing the scientific visibility of developing countries and its central role in disseminating Brazilian education research. The combined query used the terms "Education" and "Ceara" in titles, abstracts, and keywords, allowing for publications in any language. Searches were restricted to the period between 2005 and 2024, corresponding to the institutionalization of the Literacy at the Right Age Program (PAIC), which marked a pivotal moment in Ceara's reform trajectory. The initial identification phase yielded 23,434 records, distributed as follows: 14,500 in Google Scholar, 8,071 in Capes Periódicos, and 148 in SciELO. Studies unrelated to education or education policy, as well as those focusing on early childhood, kindergarten, higher education, or high school, were excluded from further consideration. After removing duplicates and applying the inclusion criteria, 166 records from Google Scholar, 528 from Capes Periódicos, and 84 from SciELO remained. Full-text screening reduced this corpus to 112 studies. The final exclusion criteria eliminated papers that did not meet the quality appraisal standards established for the review. This quality appraisal was conducted to assess the methodological robustness and relevance of the studies prior to inclusion in the final synthesis, in accordance with the PRISMA 2020 recommendations for transparent and reproducible selection processes. All studies that reached the eligibility phase (n = 112) were evaluated using a structured appraisal protocol based on established criteria for systematic reviews. The appraisal examined four core dimensions: (1) clarity of research aims, which verified whether the study articulated explicit and coherent objectives; (2) appropriateness of research design, assessing whether methodological choices aligned with the research aims and context; (3) transparency of data collection and analysis procedures, including the specification of instruments, samples, and analytical strategies; and (4) relevance of reported outcomes to learning improvements and education policies in the context of Ceara's primary and lower secondary education. Each criterion was scored dichotomously (meets/does not meet criterion), allowing for a consistent comparison across studies. Studies were excluded when they did not present sufficient methodological detail to judge validity, lacked evidence of alignment with the stated research objectives, or did not report outcomes related to learning or governance in educational settings. This appraisal resulted in the removal of 32 studies that failed to comply with a minimum threshold of methodological transparency or policy relevance, leaving 80 studies in the final analytical sample. The adoption of this appraisal procedure ensured that the synthesis was grounded in methodologically robust evidence and aligned with internationally recognized standards for systematic literature reviews. The quality appraisal followed the standardized procedures recommended by the Joanna Briggs Institute (Aromataris and Munn, 2020), ensuring that only studies demonstrating methodological rigor and adequate reporting were included in the synthesis.

Data extraction followed a structured and iterative process in which each study was examined to capture publication characteristics, methodological approaches, references to Ceara-originated instruments, governance mechanisms, and reported effects on learning outcomes. The review employed thematic synthesis complemented by metadata coding to identify analytical patterns beyond individual study findings. In the primary phase of metadata coding, the studies were categorized according to methodological approach, publication location, field of study, level of analysis, and the institutional actors involved. This phase enabled the systematic mapping of research production regarding Ceara and clarified how various academic, governmental, and community stakeholders engaged with the reform. A secondary phase of metadata coding synthesized the types of interventions associated with improved learning outcomes, generating thematic clusters that reflected the systemic and interdependent nature of Ceara's policy instruments. This approach transformed qualitative descriptions into analytical structures, allowing the review to identify convergences, temporal shifts, and interpretive patterns related to governance, accountability, and educational improvement.

Because this study relied exclusively on publicly available scholarly sources, no ethical approval was required. The retrospective protocol adopted, in conjunction with full adherence to PRISMA 2020 standards, ensured transparency, replicability, and alignment with international guidelines for systematic reviews. The analytical procedures extended beyond the identification and selection of studies and proceeded through a structured set of methods designed to extract and interpret evidence from the final corpus. The next section presents the metadata coding strategy, which constitutes a central component of the analytical methods by translating qualitative information into systematic categories and enabling the detection of recurrent patterns across policies, actors, and institutional arrangements. Building upon the coded data, descriptive analytics was applied to examine historical configurations and trace relationships among events, focusing on "descriptions of past events" that generate explanatory

insights through the use of searchable repositories (Fleckenstein and Fellows, 2018, p. 133). This integrated approach—anchored in PRISMA-based selection procedures, metadata coding, and descriptive analytics—provides a coherent methodological pathway from the identification of relevant studies to the structured interpretation of how Ceara's education reform evolved and how its policy components have been associated with reported improvements in learning outcomes.

2.1 Metadata coding

The systematic review employed thematic synthesis and meta-coding, which enabled the development of analytical themes that transcended the findings of individual studies (Voracek et al., 2019). These themes were subsequently converted into numerical codes, allowing for systematic tabulation and further analysis. During the primary meta-coding process, the studies were categorized based on general characteristics and methodological approaches. Variables included the research approach (qualitative, quantitative, or mixed methods), publication location (national, local within Ceara, or other countries), and study location (Ceara, Sobral, another municipality in Ceara, or mixed). In addition, the classification considered the field of study (such as education, public policy, economy, law, sociology, psychology, or other areas), the level of analysis (school, municipality, state, federal government, global, or multilevel), and the type of engagement (academic, government, NGO, school or local community, intersectoral, or international organization). Simultaneously, secondary meta-coding was conducted to synthesize the types of interventions associated with improved learning outcomes in Ceara. These interventions were derived inductively from the reviewed sources during the quality appraisal stage and were subsequently evaluated for their relative importance.

Ceara's reform trajectory demonstrates how a state with limited economic resources, ranked among the lowest in GDP per capita, was able to surpass wealthier states in national large-scale assessments (Instituto Nacional de Estudos e Pesquisas Educacionais Anisio Teixeira (INEP), 2023; Loureiro et al., 2020). The outcomes reveal a cumulative process in which interdependent policy instruments aligned over time to sustain improvement, a process further examined in the next section.

3 Results

3.1 Studies characteristics

The results are presented in alignment with the two research questions. To address Research Question 1, this section identifies and characterizes the academic studies that examine learning improvements in Ceara, considering their temporal distribution, methodological orientation, administrative focus, stakeholder engagement, and institutional and geographical affiliations. Of the 80 selected studies, 69 were published in Brazil, with publications concentrated mainly in Ceara, Sao Paulo, and Minas Gerais. Within the Brazilian subset, 6 studies did not specify the state of institutional affiliation, preventing precise attribution. The spatial distribution of these publications across Brazilian states is illustrated in Figure 2. The remaining 11 studies originated from outside Brazil, with publications identified in the United States (4), Portugal (1), Argentina (1), Switzerland (1), the United Kingdom (1), and India (1), as well as in broader regional outlets in Latin America (1) and Ibero-America (1). This distribution indicates that the study of learning improvements in Ceara remains predominantly situated within the Brazilian academic system, with limited but emerging international engagement.

Nonetheless, the presence of studies conducted abroad reveals a growing international interest in Ceara's education reform and its outcomes within the broader context of education policy and development. This distribution suggests that, while the reform has garnered recognition for its effectiveness within national boundaries, its conceptual frameworks and empirical findings remain less accessible to a global academic audience. The scarcity of publications in English also suggests the need for increased international dissemination and comparative analyses, which could contribute to bridging the gap between local innovation and global policy learning.

Figure 2 illustrates the annual distribution of publications from 2005 to 2024. Overall, the trend suggests a progressive expansion of research until 2020, followed by a decrease in scholarly production in subsequent years.

In terms of disciplinary focus (Table 2), 53.8% of the publications belonged to the field of Education (43 studies), indicating the predominance of pedagogical analyses. This disciplinary distribution reflects a strong alignment between the reform's educational goals and the analytical lenses used to study it. However, the relatively low representation of fields such as sociology, law, and psychology highlights the limited exploration of the reform's broader social, legal, and behavioral implications.

Regarding methodological orientation (Figure 3), the dataset reveals a clear predominance of qualitative approaches (57; 71.3%), followed by quantitative (18; 22.5%) and mixed-methods studies (5; 6.3%). This overemphasis on qualitative research reflects the interpretative nature of inquiries into educational change, which often prioritize contextual understanding, policy processes, and stakeholder perceptions over experimental or correlational analyses. Although qualitative designs have provided valuable insights into policy implementation and institutional dynamics, the limited number of quantitative and mixed-methods studies constrains the capacity to generalize findings or explore causal relationships. Consequently, future research could benefit from integrating both approaches to capture not only how the reform was implemented but also the magnitude of its effects on learning outcomes.

Analyses were conducted across various administrative levels (Figure 4). This distribution reveals that most analyses privilege macro-level interpretations, often emphasizing the role of the State Secretariat of Education as the central coordinator of reforms. The relatively few multilevel studies highlight a gap in understanding how policies are adapted and reinterpreted across governance layers, which is essential for explaining the reform's continuity and sustained results over time. Thus, the existing literature tends to conceptualize Ceara's interventions as a result of centralized state policy coordination.

Stakeholder engagement refers to the structured and sustained participation of relevant actors, including governmental agencies,



FIGURE 2
State-level distribution of selected studies in Brazil on Ceará education.

TABLE 2 Fields of study.

Field of study	Number of studies
Education	43
Public policy	20
Economy	10
Sociology	2
Law	2
Psychology	1
Administration	1
Mathematics and statistics	1
Total	80

non-governmental organizations, practitioners, and communities, in the formulation, implementation, and evaluation of public policies, to enhance legitimacy, accountability, and policy effectiveness (Bryson et al., 2013). The prevalence of state-level and institutional perspectives suggests that the literature analyzes the reform from a managerial and policy implementation standpoint, emphasizing the role of state coordination, incentives, and partnerships in sustaining improvements in learning outcomes. However, a lack of attention to community and school-level engagement signals an underrepresentation of bottom-up perspectives and the lived experiences of educators, families, and students. This imbalance may obscure critical dimensions of how educational change is internalized at the micro level and sustained through local ownership and cultural adaptation, as shown in Figure 5. The distribution of stakeholder engagement across the reviewed studies are summarized in Figure 6, indicating the actors involved in policy processes and that were the focus of the analyses.

Recognizing these patterns helps define the scope and limitations of current research, clarifying how existing analytical perspectives shape the understanding of factors driving learning improvements. Studies lacking explicit classification under these categories were not counted in the characterizations shown in Figures 4–6. However, this contextual overview informs the following section, which traces the timeline of key education policies implemented in Ceara.

3.2 Summary of key Ceara state education policies

To respond to Research Question 2, this section identifies which components of Ceara's education policies have been identified in the literature as contributing to reported improvements in learning outcomes, describing how Ceara's education reform was structured and operationalized across successive administrations. Drawing on the literature, we identify a set of state-level initiatives that were central to the consolidation of the education reform in Ceara. These programs, sustained over multiple administrations, created

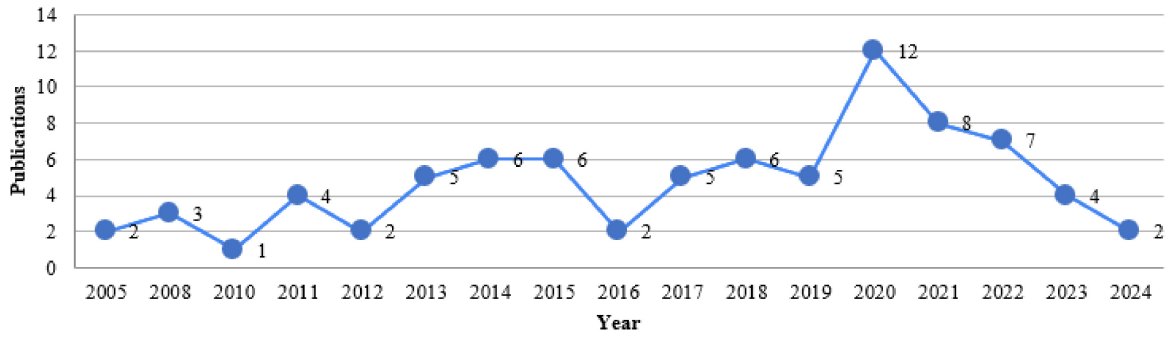


FIGURE 3
Number of publications per year.

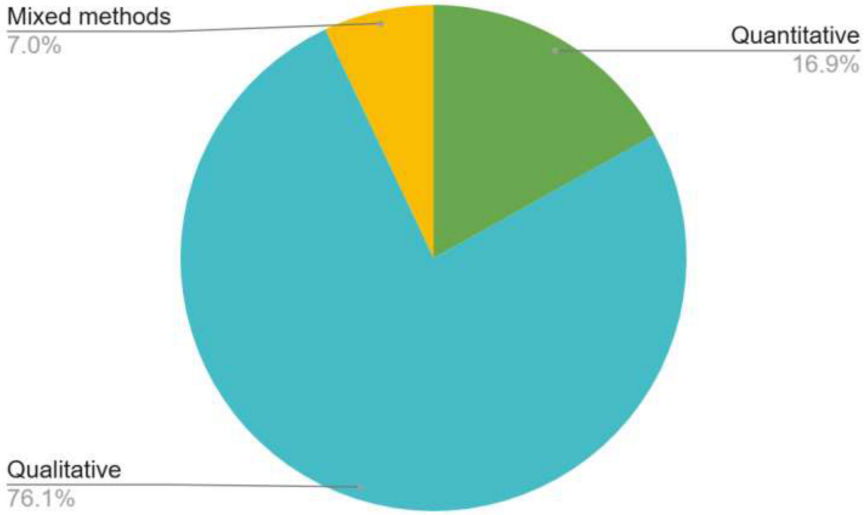


FIGURE 4
Methodological orientation.

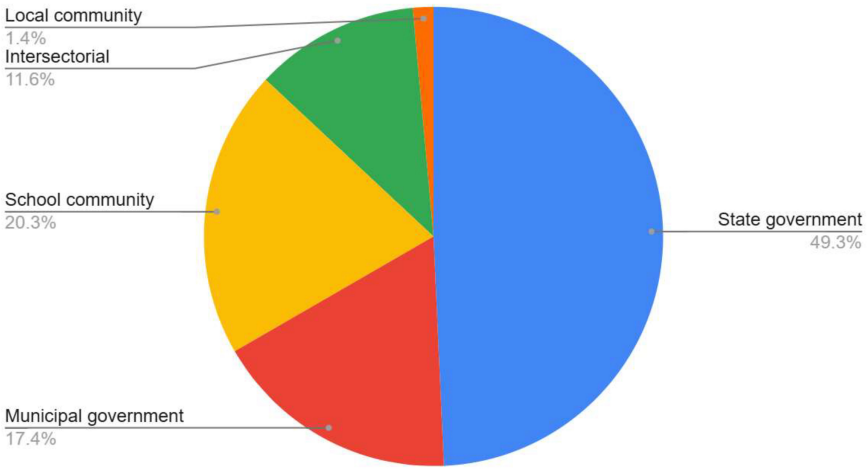


FIGURE 5
Administrative level.

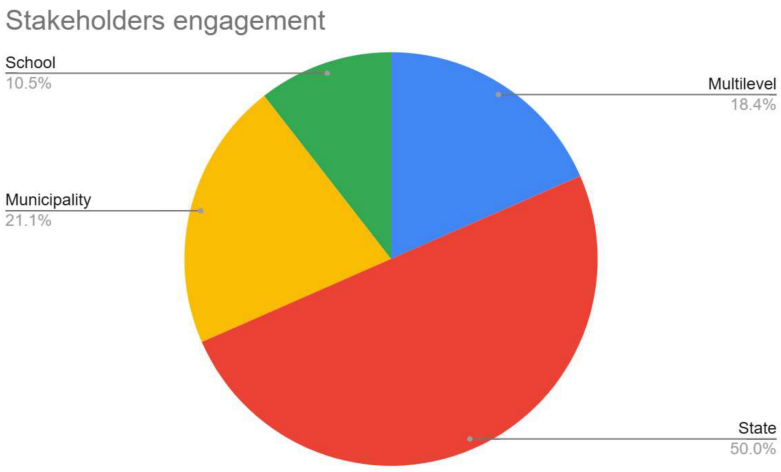


FIGURE 6
Stakeholder engagement.

a coherent architecture of evaluation, incentives, technical support, and redistribution that together helped improve learning outcomes. The following is a narrative of those initiatives, along with a brief description of each.

The first significant step was the creation of the Permanent System of Evaluation for Basic Education (SPAECE) in 1992, a state-wide, large-scale external assessment system (Carvalho M. S. B., 2014; da Silva, 2014; Soares and Werle, 2016). SPAECE typically tests Portuguese and mathematics for students in fundamental and secondary schooling and includes contextual questionnaires of students, teachers, and school management. Over time, the system acquired multiple strands, such as SPAECE Alfa for 2nd-grade literacy, and became not only a measuring device but also a diagnostic tool for policy, enabling the state and municipalities to monitor progress and guide pedagogical strategies.

In Brazil, although the Federal Government was formally assigned a more coordinating role, little progress had been made toward establishing effective mechanisms until 1995. During the administrations of Fernando Henrique Cardoso (1995–2002) and Luiz Inácio Lula da Silva (2003–2010), federal coordination expanded significantly through the approval of national guidelines and regulations, the redistribution of financial resources, and the promotion of federal education programs (Segatto, 2015). In the same period, Ceara, which in the 1990s had some of the lowest literacy rates in Brazil, redefined its education framework to strengthen the institutional and managerial capacities of central, regional, municipal, and school levels, thereby assigning municipalities greater responsibility for delivering basic education services (de Gusmão and Ribeiro, 2011; Souza and Frota, 2011; Ramos and Soares, 2012). This reform transferred more operational control over schooling to municipal secretariats, making local governments more directly accountable for learning outcomes in the early grades. As part of this process, the state enacted Law No. 12,452 to regulate the municipalization of public education within the framework of the Sustainable Development Plan and the "Todos pela Educação de Qualidade para Todos" (Education of Quality for All) program. The State Secretariat of Education (SEDUC) established a working group to design mechanisms for technical and financial cooperation. In October 1995, an agreement was signed between the state and six municipalities (Icapuí, Fortim, Maranguape, Marco, Iguatu, and Jucás) to transfer responsibility for the early grades of primary education to municipal governments. The initiative led to the creation of Fundefinho in 1997, through which the state signed agreements with 124 municipalities that assumed responsibility for the early grades, setting a minimum per-student cost of R$180.00 (Vieira, 2010).

The term Fundefinho, a diminutive of FUNDEF, refers to the Fundo de Manutenção e Desenvolvimento do Ensino Fundamental e de Valorização do Magistério (Fund for the Maintenance and Development of Primary Education and the Valorization of Teachers). The program operated as a pilot for the later implementation of FUNDEF at the national level, in partnership with the Ministry of Education. According to Vieira (2010), the transfer of funds from the state to municipalities through Fundefinho and later FUNDEF stimulated a rapid municipalization process that diverged from SEDUC's original design. Because FUNDEF provided additional financial resources to local administrations, municipalities expanded their control over the entire primary education system rather than only the early grades (Vieira, 2010).

Between 1995 and 2002, this process also encouraged new forms of intergovernmental cooperation, including a unified teacher recruitment exam organized by SEDUC in partnership with 153 municipalities, the Programa Permanente de Apoio aos Secretários Municipais de Educação e Prefeituras (Permanent Support Program for Municipal Education Secretaries and City Halls), and in-service teacher training programs developed in collaboration with four public universities in Ceara (Naspolini, 2001; Vieira, 2010; Segatto, 2015).

The political prioritization of literacy gained strength in 2004, with the creation of the Cearense Committee for the Elimination of School Illiteracy by the Legislative Assembly. Coordinated with the participation of universities and civil society institutions, the committee conducted a study to assess literacy levels among second-grade students, finding that only 15% were literate. This

alarming result mobilized diverse actors and encouraged the formation of a network of partners to seek and test solutions (de Gusmão and Ribeiro, 2011; Becskeházy, 2018).

From this articulation, the Literacy at the Right Age Program (PAIC) (Ceara, 2015b) was established to support municipalities in reducing illiteracy in the early years of primary education. Its initial phase took place between 2005 and 2006. It involved 56 of the 184 municipalities in the state, under the coordination of the Association of Municipalities and Mayors of the State of Ceara (APRECE) and the Union of Municipal Education Leaders of Ceara (UNDIME-CE), in partnership with UNICEF (Ceara, 2015b). During this pilot stage, the program's main activity consisted of conducting diagnostic assessments of students' reading and writing skills (de Gusmão and Ribeiro, 2011). This phase also served to validate assessment instruments, teacher training methodologies, strategies for state–municipal cooperation, and technical coordination mechanisms. Following positive results and adjustments, the program was institutionalized and expanded to all municipalities in the state through Law No. 14,026 of 2007, along with formal cooperation protocols with municipal governments (Ceara, 2007).

In 2007, the State Department of Education (SEDUC) assumed direct coordination of PAIC. The program targeted the first and second years of primary education. It was structured around five axes: municipal education management, external evaluation, literacy teaching, reading development, and early childhood education. Actions within these axes included training workshops, technical support both remotely and in person, and the provision of teaching instruments and materials, with reinforced support for municipalities identified as priorities (Ceara, 2012). All municipalities in the state adhered to the program, facilitated by the near-universal municipalization of primary education in Ceara at the time. The first actions were effectively implemented in 2008, including the training of literacy teacher-trainers for first-grade students, direct training of literacy teachers in second grade, and the production and distribution of structured teaching materials for both years (Ceara, 2015a). The design of PAIC was grounded in a collaborative model between the state and municipalities, characterized by shared responsibility, mutual accountability, and technical cooperation. To operationalize this collaboration, SEDUC established the Coordenadoria de Cooperação com os Municípios (COPEM) and a network of regional coordination offices (CREDES) to provide technical support to local education teams (Souza and Frota, 2011). Municipalities joined the program by signing formal agreements and creating local teams responsible for implementing the actions across the five program axes. This structure allowed the state to respect municipal autonomy while ensuring technical coherence and alignment with the broader state education policy (de Gusmão and Ribeiro, 2011; Secretaria da Educação do Estado do Ceara [SEDUC], 2012).

Evidence from Costa and Carnoy (2015) suggests that PAIC bears resemblance to the Reading First program implemented in the United States, particularly in its emphasis on structured literacy materials and teacher training. However, there were important differences. Unlike Reading First, which operated with a loose connection between federal funding, states, and districts, PAIC was centrally administered by SEDUC. The state developed literacy materials, organized teacher training workshops across 23 centers, and conducted in-person visits to support implementation. This direct role allowed more substantial alignment between policy design and classroom practice. Moreover, PAIC combined teacher training with broader instructional improvement and fostered a culture of networked collaboration, whereby municipalities and schools shared best practices for literacy development. Notably, the program introduced financial incentives for municipalities and schools whose students demonstrated the largest literacy gains in the Space Alfa assessment, a mechanism not present in the U.S. program. Given the lower initial achievement levels in Brazil compared to the United States, Costa and Carnoy (2015) argue that PAIC's design and context allowed for stronger effects on learning outcomes than might be expected from similar interventions in high-income countries.

Also in 2007, a new amendment to state law introduced a results-based redistribution of state tax (ICMS) revenue, tying part of municipal transfers to improvements in learning outcomes. Municipalities that achieved better performance in state assessments received higher shares of revenue, creating fiscal incentives to prioritize education. Another financial incentive mechanism consistent with this approach was the Escola Nota Dez Award, established in 2009 through Law No. 14.371. The award recognizes 150 public schools with the highest literacy results based on the School Performance Index—Literacy (IDE-Alfa). Each awarded school receives financial support per student in two installments, 75 and 25%, contingent upon submitting a spending plan and establishing a technical-pedagogical cooperation agreement with one of the 150 lowest-performing schools. Supported schools receive half the value per student, also divided into two installments. The second payment depends on the accountability compliance, maintenance, or improvement of results by the awarded school, as well as progress by the supported school (Secretaria da Educação do Estado do Ceara [SEDUC], 2012).

The initiative seeks to value school performance and encourage improvement through collaboration and knowledge exchange. It aimed to promote the dissemination of management and pedagogical practices focused on learning outcomes. As noted by Abrucio et al. (2017), this arrangement combines incentive-based competition with cooperative mechanisms between schools, a distinctive feature of Ceara's collaborative governance model. The Escola Nota Dez (Top Ten School Award) differs from other incentive-based education policies implemented in Brazil due to its combination of two contrasting principles. On one side, it incorporates a meritocratic orientation by rewarding municipalities and schools that achieve the best results, thereby encouraging accountability and performance-based competition.

On the other hand, it integrates an equity-driven approach by providing technical and financial support to low-performing schools, aiming to reduce disparities in learning outcomes and promote adequate achievement levels among schools serving socially disadvantaged students (Muylaert et al., 2022). Over time, the award's criteria have been revised to enhance fairness and collaboration. The updated regulations introduced prerequisites such as requiring awarded schools to support those facing difficulties, considering the number of enrolled students, and excluding winning schools from the following year's competition. Additionally, adjustments were made to the calculation of average performance and learning factors to prevent the system from favoring schools with a large proportion of low-proficiency

students (Secretaria da Educação do Estado do Ceara [SEDUC], 2012; Abrucio et al., 2017; Muylaert et al., 2022).

The scope of the reform expanded in 2011, when the government extended PAIC to the 5th grade of elementary school through PAIC + 5, reinforcing continuous learning gains beyond literacy in Portuguese, mathematics, and other foundational competencies. In 2015, Mais PAIC (PAIC Plus) further broadened the program to support students from 6th to 9th grades, sustaining progress achieved in earlier stages and preparing them for secondary education (Ceara, 2025). Collectively, these state-level initiatives, which included external evaluation through SPAECE, municipal responsibility for primary education, the legislative committee on illiteracy, and incentive mechanisms such as ICMS redistribution, constituted a mutually reinforcing policy ecosystem aimed at improving learning outcomes throughout the basic education cycle.

Despite Ceara's political turnover, with 14 governors from six parties and 16 different education secretaries since Brazil's redemocratization in 1986, these policies persisted, were refined, and in several cases expanded. The literature suggests that this continuity reflects a shared commitment across levels to pedagogical alignment, the desire for revenue bonuses and grants tied to results, and accountability mechanisms that anchored these programs in the state's educational apparatus.

3.3 Tracing the understandings of the improved learning outcomes

Continuing the response to Research Question 2, this section examines how the literature has interpreted and redefined the relationship between Ceara's policy mechanisms and reported improvements in learning outcomes over time, highlighting shifts in the conceptualization of educational quality. The systematic review of studies published between 2005 and 2024 reveals a change in how scholars have conceptualized and examined the improvement of learning outcomes in Ceara. Between 2005 and 2012, research focused on developing and institutionalizing large-scale assessment and monitoring systems as instruments to ensure accountability and strengthen education governance. During this period, educational quality was defined mainly by the capacity of state institutions to measure, monitor, and regulate school performance. Studies by Lima and Andrade (2013), Pequeno et al. (2010), and Lima (2012) examined the development of permanent systems of educational assessment and the use of regulatory mechanisms to enhance administrative efficiency. Overall, this phase reflected a governance approach in which the existence of evaluation structures and managerial arrangements was considered a central indicator of educational quality.

In transition, from 2011, the focus shifted toward evaluating the outcomes of these systems, especially regarding early grade literacy (de Gusmão and Ribeiro, 2011; Vieira and Vidal, 2011; Sousa, 2013; da Silva, 2014). Studies such as those by de Gusmão and Ribeiro (2011) have emphasized the role of collaboration between state and municipal governments in achieving literacy at the right age, marking the consolidation of the link between assessment mechanisms and measurable learning gains. Educational quality came to be defined less by institutional design and more by its tangible effects on student performance, particularly in the early years of schooling.

Beginning in 2013, the concepts of equity and inclusion were introduced into the analysis of educational quality (Padilha et al., 2013). Padilha et al. (2013) explicitly linked quality and equity in their examination of fundamental education, signaling a transition from evaluating average performance to analyzing its distribution across social groups. This perspective established equity as an intrinsic dimension of quality.

From 2017 onward, research diversified and deepened along three main lines (for instance, Kasmirski et al., 2017; Passone and Araújo, 2020). First, assessment and accountability, once seen as innovations, became institutionalized as part of a broader management infrastructure. Second, governance and coordination analyses expanded to include implementation routines and institutional capacity, highlighting that sustained improvement depends on collaboration and technical expertise across administrative levels. Third, the literature adopted a distributive perspective, examining how educational gains are shared among different populations. Finally, recent studies have focused on specific policy instruments, such as financial incentives and award-based mechanisms. Ponne (2023), for example, applied quasi-experimental methods, including synthetic control, to estimate their causal impact on learning outcomes.

3.4 Interventions associated with the improved learning outcomes

Further advancing the response to Research Question 2, this section organizes the interventions identified in the reviewed studies into thematic clusters, clarifying how different policy instruments and implementation strategies have been associated with learning improvements in Ceara's primary and lower secondary education. The review of the selected studies reveals the understanding of the types of interventions associated with improved learning outcomes in Ceara's primary and lower secondary education. Early research focused on foundational policy instruments, including redistributive financing mechanisms, large-scale student assessments, and early literacy programs. These measures reflected the consolidation of a results-based management approach that emphasized performance monitoring and accountability. Over time, the focus expanded to include governance arrangements, intersectoral collaboration, and professionalization of management as essential for sustaining progress in educational quality.

To better understand these developments, the identified interventions were reorganized into thematic clusters, reflecting the systemic and interdependent nature of Ceara's education reform.

3.4.1 Intersectoral actions and education programs

The articulation between intersectoral collaboration and education programs formed the foundation of Ceara's reform. Arnold and Sousa (2011) and de Gusmão and Ribeiro (2011) emphasized that strategic management and the collaborative governance established between the state and municipalities allowed for policy alignment and sustained implementation.

Gadelha and Frota (2013) and Carvalho M. S. B. (2014) further noted that the consolidation of integrated management instruments and the continuity of large-scale assessments, such as SPAECE, reinforced program consistency across political cycles. Abrucio et al. (2017) demonstrated that Ceara's regime of collaboration became a central feature of its governance structure, fostering institutional resilience and mitigating the disruptive effects of political transitions.

Building on this foundation, more recent studies have shown how this alignment evolved into a durable, system-wide model. Becskeházy (2018) and Mota et al. (2019) argued that the institutionalization of collaborative practices effectively transformed the pursuit of educational quality into a state policy. Vieira et al. (2019) emphasized that coherence among planning, assessment, and funding instruments underpins the sustainability of results-based management. Coordination across federal, state, and municipal levels enabled continuity of the PAIC, which sought to guarantee foundational literacy during primary education. In this sense, Costa and Ramos (2020) described the PAIC collaboration framework as a mechanism that combined local autonomy with shared accountability, ensuring continuity amid administrative change. Similarly, Loureiro et al. (2020) concluded that Ceara's long-term success in reducing learning poverty resulted from the alignment of policies and the enduring intergovernmental coordination that preserved coherence over time. Collectively, these studies indicate that Ceara's educational advances were not merely programmatic but institutional, grounded by a governance model that reconciled decentralization, accountability, and policy stability.

On the other hand, Vidal and Vieira (2014) analyzed municipal education policies under accountability frameworks. They found that managerial control tends to constrain teachers' autonomy and narrow curricular focus to test-oriented skills. Cavalcante (2014) emphasized the need to reconcile managerial efficiency with democratic school management, warning that performance-based governance may erode participatory decision-making.

3.4.2 Assessment and financing mechanisms

The integration of large-scale student assessments with redistributive financing mechanisms established the accountability and incentive structure of the reform. As previously mentioned, the SPAECE system is a state-level initiative designed to evaluate education locally, providing data tailored to regional needs that goes beyond what is offered by national assessment processes. It provided regular evaluations whose data guided school planning and resource allocation. At the same time, the redistribution of part of the state's ICMS tax revenues according to municipal performance encouraged local governments to prioritize education. This connection between evaluation and funding created a self-reinforcing process of monitoring and improvement, characteristic of Ceara's results-based governance model.

Performance-based financing and large-scale assessments, central to Ceara's model, also pose risks of increasing competition and inequality (Pessoa and dos Santos, 2020; Zientarski et al., 2019). da Silva (2014) examined how school managers internalize and use SPAECE assessment results, arguing that the emphasis on quantifiable performance indicators can lead to symbolic control over pedagogical practices and shift teachers' attention from learning processes to measurable outcomes. Similarly, Carvalho M. S. B. (2014) highlighted both the potential and the limitations of SPAECE as a monitoring tool, noting that its technocratic orientation risks marginalizing creative and civic dimensions of learning. In parallel, Brandão (2014) analyzed the redistribution of the ICMS tax based on educational performance and concluded that, while effective in fostering accountability, such mechanisms can inadvertently reinforce inequality and instrumentalize the meaning of educational quality. Collectively, these studies underscore that Ceara's success in improving learning outcomes coexists with tensions between managerial rationality and the broader humanistic aims of education. Although Ceara mitigated these through pedagogical support, sustainability remains uncertain. Attempts to replicate the model in Maranhão, Espírito Santo, and Sergipe have yielded uneven results (Todos pela Educação, 2021), emphasizing the importance of governance capacity and local conditions.

3.4.3 Participatory management

A strong participatory and collaborative orientation characterized the implementation of Ceara's education reform. Arnold and Sousa (2011) highlighted that the state's strategic planning approach relied on coordination among school managers, local authorities, and community actors to achieve shared educational goals. de Gusmão and Ribeiro (2011) emphasized the intergovernmental collaboration underpinning the Programa Alfabetização na Idade Certa (PAIC), in which state and municipal administrations worked alongside organizations such as UNICEF and UNDIME to sustain collective engagement and capacity building. Ramos et al. (2012) noted that the modernization of Ceara's education governance was supported by cooperative networks that enhanced policy continuity and alignment across administrative levels. Gadelha and Frota (2013) emphasized that the program's effectiveness was partly due to community participation and school-level collaboration, which reinforced accountability and commitment to educational improvement. Complementing these findings, Costa and Carnoy (2015) demonstrated that family and community involvement played a significant role in enhancing literacy outcomes, suggesting that civic engagement amplified the effects of results-based management. Together, these studies indicate that Ceara's model integrated managerial efficiency with participatory governance, promoting continuity and social legitimacy in the reform process. Over time, participatory governance has become increasingly visible, indicating a shift from centralized decision-making to shared responsibility and joint accountability among stakeholders at both state and municipal levels.

3.4.4 Recruitment and professionalization of teachers and principals

The institutionalization of merit-based recruitment and professionalization marked a turning point in Ceara's education reform, redefining school leadership and teaching careers within a framework of accountability and results orientation. Early studies documented the initial shift from politically driven appointments to technically based selection procedures. Arnold and Sousa (2011) described how strategic management reforms introduced public calls for school principals, requiring written exams, evaluation of titles, and management plans, thereby

replacing partisan appointments with merit-based selection procedures. In line with this perspective, de Gusmão and Ribeiro (2011) noted that municipal governments progressively adopted professional recruitment. Subsequent research highlighted how these mechanisms change to a more comprehensive model. Vidal and Vieira (2014) analyzed how municipal education systems adapted to accountability pressures by institutionalizing technical standards for recruitment and performance evaluation. Finally, Costa (2016) observed that under the state's intergovernmental collaboration framework, professionalized recruitment and managerial leadership became central to sustaining institutional stability and a merit-based culture across municipalities.

Some analyses suggest that this meritocratic and professional framework has become increasingly institutionalized and locally adapted across municipalities. Becskeházy (2018) highlighted that the state's governance arrangements, centered on professional leadership and technical selection, contributed to the consolidation of education as a state policy rather than a government policy. Vieira et al. (2019) observed that the institutionalization of recruitment processes and leadership training strengthened managerial coherence and continuity amid political transitions. Loureiro et al. (2020) emphasized that Ceara's success in sustaining learning improvements was closely associated with merit-based appointments and professionalized school management, which ensured the consistent implementation of education policies. A more recent municipal-level study shows that competitive and transparent selection processes for principals, based on technical qualifications, written assessments, and school management plans, have become the standard practice, replacing politically influenced appointments (Brito et al., 2024). Together, these studies reveal that Ceara progressively institutionalized a results-based and merit-oriented education system, rooted in professional competence, leadership accountability, and sustained policy coherence.

3.4.5 Capacity-building and school-level actions

Capacity building and school-level actions remained essential components of Ceara's education reform. In the early stages, initiatives focused on developing managerial and pedagogical capacity through technical training and leadership development. Souza and Frota (2011) and Arnold and Sousa (2011) described how regional support centers (CREDES) and state programs provided in-service training and technical assistance to align pedagogical and administrative practices. de Gusmão and Ribeiro (2011) and Lima (2012) examined how training combined with diagnostic feedback from assessments supported teachers in using learning data to guide instruction. Shirasu et al. (2013) observed that financial incentives and performance-based rewards reinforced this professional development framework, strengthening a culture of results-oriented practice. Costa and Carnoy (2015) noted that professional training contributed to improving early grade literacy, while Costa (2016) linked the institutionalization of leadership programs to the intergovernmental collaboration framework. More recent analyses, such as those by Maciel (2020) and Ribeiro et al. (2020), have identified continuing education and technical assistance as mechanisms for ensuring equity and consistency in teaching practices. Similarly, Loureiro and Cruz (2020) and Loureiro et al. (2020) described how structured training and principal development were central to the Sobral and Ceara models, reinforcing the integration between professional learning and policy implementation. Andrade et al. (2021) also demonstrated that teacher training has increasingly incorporated assessment data from SPAECE, thereby promoting evidence-based pedagogical practices.

At the school level, the reform combined capacity building with organizational and curricular adjustments that strengthened instructional coherence. Gadelha and Frota (2013) observed that curriculum planning, school management improvement, and the expansion of full-time schools contributed to greater alignment between classroom practices and state goals, while for and Freire et al. (2020) the consolidation of SPAECE enhanced system monitoring, learning outcomes and teacher accountability. Vidal and Vieira (2014) and Vieira et al. (2019) discussed how municipalities incorporated pedagogical supervision and school-based leadership development to consolidate the outcomes of reform. Recent studies by Loureiro and Cruz (2020) and Loureiro et al. (2020) indicated that local implementation in Sobral and other municipalities sustained coherence among teacher support, school management, and student learning. Altogether, these analyses suggest that the professional learning system and school-level strategies evolved in parallel, linking capacity development, pedagogical alignment, and institutional continuity over time.

3.4.6 Support through teaching materials, infrastructure, and contextual factors

Improvements in school materials, infrastructure, and pedagogical resources, together with attention to social and territorial disparities, have shaped Ceara's broader learning environment. Initial research highlighted the relevance of material conditions and regional inequalities for understanding educational effectiveness. Vieira et al. (2005) analyzed the relationship between per-student spending and regional disparities, suggesting that unequal access to infrastructure and pedagogical resources limited school performance. Likewise, Souza and Frota (2011) and Arnold and Sousa (2011) noted that investments in educational facilities, teaching materials, and logistical support through the CREDES network strengthened local school capacity and improved management practices. Gadelha and Frota (2013) observed that infrastructure enhancement and resource allocation were linked to better student outcomes, while de Gusmão and Ribeiro (2011) emphasized that intergovernmental collaboration was instrumental in reducing regional inequalities through technical and material support.

Further contributions reinforced the importance of structural and contextual dimensions in sustaining learning improvements. Vidal and Vieira (2014) highlighted that municipal education systems faced persistent disparities in infrastructure and administrative capacity, which shaped how accountability-oriented policies were locally implemented. Costa and Carnoy (2015) found that school-level resources and the equitable distribution of financial and pedagogical support mediated early grade literacy gains under the PAIC. Crisóstomo and Martins (2019) extended this discussion by showing that educational performance remained closely associated with socioeconomic and territorial inequalities, revealing the interdependence between education policy and broader development factors. Building on these insights, Vieira et al. (2019) argued that Ceara's results reflected a multidimensional governance framework integrating financial

redistribution, pedagogical assistance, and social coordination. Loureiro and Cruz (2020) further demonstrated that even in adverse socioeconomic contexts, the combination of material investment, managerial coherence, and intersectoral collaboration enabled sustained improvements in student learning.

Although the reviewed studies differ in methodological focus, they consistently describe Ceara's reform as a reference for improving learning outcomes, providing an empirical basis for the conclusions that follow. Overall, the evidence suggests that Ceara's educational progress resulted from the convergence of multiple interdependent factors rather than from isolated interventions. This multidimensional pattern provides a foundation for understanding how Ceara's coherence across governance levels leads to improved learning outcomes, an issue further explored in the subsequent discussion on policy integration and institutional design.

4 Discussion

Building on the findings presented in response to the research questions, this section discusses how the characterization of Ceara's education reform model and the policy components reported as influencing learning outcomes intersect with broader debates on educational governance, implementation capacity, and subnational policy dynamics in Brazil. The synthesis of the reviewed studies shows that Ceara's education reform was not the result of a single innovation but of the progressive articulation of multiple, interdependent factors. The six clusters identified, intersectoral collaboration and education programs, assessment and financing mechanisms, implementation and participatory management, recruitment and professionalization of educators, capacity-building and school-level actions, and contextual and infrastructural improvements, together suggest that policy alignment across governance, pedagogical, and structural dimensions may have contributed to sustaining long-term improvements in learning outcomes in primary and lower secondary education. In line with Andrews et al.'s (2017) notion of adaptive state capability, the state's model demonstrates that stable intergovernmental collaboration and iterative policy learning can compensate for resource constraints and political volatility.

However, the review also reveals structural and ethical challenges that future research and policy design should consider. The association between performance and funding, though effective in promoting results, generated competitive pressures among municipalities and schools. Some studies report that this dynamic encouraged the exclusion of students with disabilities during testing days, reinforced the homogenization of teaching, and led to practices oriented primarily toward standardized test preparation rather than comprehensive learning (Carvalho M. S. B., 2014; Passone and Araújo, 2020). Others point to the risk of managerial control and the reduction of educational goals to measurable indicators, potentially constraining schools' autonomy and their capacity to work on qualitative dimensions such as creativity, citizenship, and inclusion. Arnold and Sousa (2011) observed that the adoption of strategic management instruments introduced corporate-style control mechanisms that could limit pedagogical flexibility. de Gusmão and Ribeiro (2011) similarly highlighted that the close collaboration between the state and municipalities, though effective for coordinating targets, sometimes restricted local autonomy in defining qualitative educational priorities. Lima (2012) argued that the growing centrality of large-scale assessments risked reducing the educational agenda to what can be measured, thereby overlooking dimensions such as creativity and citizenship. Ramos et al. (2012) noted that managerial modernization of education governance reinforced a culture of performance monitoring, potentially subordinating educational goals to bureaucratic efficiency. Similarly, Gadelha and Frota (2013) noted that an emphasis on measurable outcomes may limit schools' capacity to promote inclusion and holistic learning experiences. These findings suggest that accountability mechanisms must be balanced with pedagogical support and safeguards against exclusionary effects.

Beyond these achievements, several studies highlight unintended consequences of Ceara's education reform that warrant careful consideration in future analyses. As Luckesi (2011) cautions, when assessment becomes detached from the teaching-learning process, it risks losing its formative role and reducing education to a competition for scores rather than a process of human development. The strong emphasis on standardized testing has led to curricular narrowing and homogenized teaching practices, prioritizing measurable outcomes over broader educational aims, such as citizenship and critical thinking (Carvalho M. (2014)). While programs such as PAIC have improved learning outcomes in Portuguese and mathematics, they have not substantially reduced disparities originating in early childhood education, underscoring the need for stronger integration across educational stages (Costa and Carnoy, 2015). More recent studies have identified the exclusion of students with disabilities from testing days, leading to internal forms of exclusion (Passone and Araújo, 2020), as well as the transformation of accountability mechanisms into instruments of control focused on efficiency, competition, and performance-based funding (Maciel, 2020; Pessoa and dos Santos, 2020).

This paper highlights that Ceara's achievements resulted from its ability to institutionalize collaboration as a governance norm, rather than as a temporary arrangement. This finding extends previous discussions on state capacity by emphasizing that durable improvements in education are likely to emerge when administrative, financial, and pedagogical mechanisms operate in a mutually reinforcing manner (Fullan, 2019; Andrews et al., 2017; OECD, 2018; World Bank, 2018).

By consolidating previously dispersed empirical findings, this review advances a more coherent understanding of how Ceara's reform model has been interpreted and operationalized elsewhere, clarifying both the opportunities and limitations of policy borrowing as a strategy for improving learning outcomes in decentralized education systems. At the same time, Ceara's model contributes to global debates on results-based governance and policy coherence. While cross-country analyses have demonstrated that accountability systems and teacher professionalization can improve learning outcomes through performance incentives and monitoring (Bruns and Luque, 2015; Angrist et al., 2023), Ceara's experience represents a different configuration. It shows that coherence, understood as the alignment of evaluation, financing, and implementation, requires continuous institutional adaptation,

long-term commitment, and collaborative forms of governance. In this context, Ceara's reform was characterized by a sustained process of coordination between the state and municipalities, combining accountability with capacity-building and redistributive support.

5 Limitations and future research

Existing studies on Ceara's reform have largely focused on state and municipal management, offering limited insights into how school-level practices, teacher beliefs, and community engagement translate policy into learning outcomes. The concentration on literacy and numeracy also narrows the understanding of educational quality, often equating improvement with standardized test performance while overlooking non-cognitive and inclusive dimensions.

Moreover, although Ceara's experience has inspired federal and state-level programs in Brazil (Brazil, 2013, 2023; Todos pela Educação, 2021), the varied outcomes of these initiatives indicate that policy borrowing requires careful adaptation to local institutional and political contexts. Future research should therefore examine the mechanisms that mediate such transfers, exploring how governance capacity, political continuity, and social participation shape the success or limitations of borrowed policies. Comparative and longitudinal studies that integrate qualitative and quantitative evidence, linking accountability, equity, and broader curricular goals such as Education for Sustainable Development, could enhance understanding of how subnational reforms contribute to sustainable and context-sensitive educational improvement.

Author contributions

AMNT: Writing – review & editing, Writing – original draft.

Funding

The author(s) declared that financial support was received for this work and/or its publication. This research was partly supported by the Japanese Government (MEXT) Scholarship Program, which provides monthly financial support for doctoral studies at the University of Tokyo. The funder had no role in the study design, data collection and analysis, decision to publish, or preparation of the manuscript. This research received no other specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Acknowledgments

Yuto Kitamura contributed to revising the manuscript. English grammar and stylistic refinement were supported by OpenAI's ChatGPT (model GPT-5, developed by OpenAI, San Francisco, CA, United States).

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was used in the creation of this manuscript. English grammar and stylistic refinement were supported by OpenAI's ChatGPT (model GPT-5, developed by OpenAI, San Francisco, CA, United States).

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